



RESEARCH ARTICLE

Potato leaf disease detection using PhytoResLSTM: A hybrid deep learning approach for agricultural sustainability

Anita Rani Mehta¹, Pardeep Kumar¹, Sanjay Tyagi¹ & Shalini Aggarwal²

¹Department of Computer Science and Applications, Kurukshetra University, Kurukshetra 136 119, Haryana, India

²Shaheed Udham Singh (SUS) Government College, Matak Majri, Indri, Karnal 132 041, Haryana, India

*Correspondence email - anita_mehta.dcsa@kuk.ac.in

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Abstract

Diagnosis of potato leaf diseases is an important factor in ensuring the sustainability of agriculture. The paper introduces a hybrid deep learning approach (PhytoResLSTM) to classify potato leaf diseases based on ResNet50 and BiLSTM architectures. In this approach, ResNet50 was used for deep feature extraction and bidirectional long short-term memory (BiLSTM) network to capture the long-term relationships of sequential data to improve disease classification accuracy. The proposed method was tested using PlantVillage dataset that containing three classes: late blight, early blight and healthy leaf images. The metrics used in evaluation such as accuracy, precision, recall and F1-score show that the model achieved 98.61 % accuracy in classifying leaf disease. The proposed technique with precision of 94.62 % and recall of 97.96 % validates early detection of the leaf diseases, where false positives and false negatives are reduced to a minimum. An F1-score of 96.15 % represents an ideal balance between accuracy and error reduction. The model perfectly identified the early and late blight with the area under the curve (AUC) of 1.00 and 0.97 for healthy leaf with an overall ROC-AUC of 0.98 exhibiting excellent classification. The confusion matrix further confirms that the model demonstrates excellent performance, characterised by minimal misclassification in the identification of the potato leaf diseases. According to the state-of-the-art comparison the proposed hybrid model outperforms conventional feature extraction and leaf disease classification methods underscoring the strength of hybrid deep learning methods and suggesting their application to improve food security and sustainable agricultural systems.

Keywords: accuracy; BiLSTM; hybrid approach; leaf diseases; PlantVillage dataset; potato; ResNet50

Introduction

In India, agriculture is the major source of livelihood for about 58 % of the 1.38 billion population (1). The gross cropped area under food grains crops was reduced from 73 % to 63 % from the year 1982–83 to 2016–17 and shifted towards more profitable crops including fruits, oilseeds and vegetables (2). It is important to know that vegetables can potentially mitigate malnutrition and micronutrient deficiency especially in children and women and the vegetable production increased 10.45 times from 19.1 million tons (Mt) in 1961–63 to 200.45 Mt in 2021–22 (3, 4). According to the food and nutritional security, potato (*Solanum tuberosum* L.) ranks third among agricultural crops after rice and wheat. It is cultivated on 2.14 million ha with productivity and production of 24 t ha⁻¹ and 51.31 Mt respectively in India (5). About 90 % of potato production is contributed by six states including West Bengal, Uttar Pradesh, Gujarat, Madhya Pradesh, Bihar and Punjab. However, more than 70 % is produced by Uttar Pradesh (33.46 %), West Bengal (24.12) and Bihar (15.09 %) (6).

Potato is cultivated in different agro-climates and is susceptible to severe biotic and abiotic stresses causing substantial yield losses due to crop failure (7). The biotic stresses including the causative pathogens such as fungus, bacterial, viral,

nematodes, mycoplasma and insect-pests are responsible for the occurrence of diseases in potato (8). The stress due to high temperature, nutrient deficiency, water deficit, salinity and frost are abiotic stresses (9). Among all the biotic factors, early blight and late blight induced by the fungi, *Alternaria solani* and oomycete, *Phytophthora infestans* respectively are the most important leaf illnesses that seriously affect the productivity of potato (10, 11). Consequently, early blight can reduce the tuber yield by 50–75 %, whereas late blight is highly epidemic and can cause 100 % yield loss in potato crop (12, 13). *Phytophthora infestans* has the potential to change its pathogenic pace and high survivability under different climatic conditions leading to challenging treatment of late blight (14). However, the selection of disease resistant cultivars is the way to restrict the spread of late blight but limited availability of quality seed material of resistant cultivars is a critical issue for farmers. Chemical control is commonly followed by farmers for early blight and late blight management (15).

Early blight initially appears in irregular shapes usually with concentric rings of dark brown to black spots on the leaves of potatoes. The late blight presents as a pale green colour on olive-green lesions that spread like water soaked and brown-black on the infected leaves of potato (16). Diagnosis and identification of potato leaf diseases is of great importance in ensuring sustainable

crop production especially with the increased agricultural demands. The conventional methods of detecting plant diseases can be carried out through the application of various direct and indirect methods, which involves visual analysis, culturing and plating, immunology-based, polymerase chain reaction (PCR), fluorescence *in situ* hybridisation (FISH), volatile organic compounds (VOC) and biosensors (17). The traditional methods of identifying the leaf diseases of plant are typically laborious and prone to error particularly where the symptoms are less severe or where they are at the early stages of infection (18). In the recent past, machine learning (ML) and deep learning (DL) have gained popularity as one of the most effective solutions to the improved accuracy, efficiency and scalability of leaf illness detection. The ML models can process large amounts of visual images of plants and extract complicated designs that often cannot be detected by humans. Support vector machines (SVM), k-nearest neighbours (KNN), logistic regression (LR) and random forest (RF) were also successfully utilised as ML models to predict leaf illness in potato based on its colour, texture and shape as the distinguishing characteristic of different types of infections, such as late blight and early blight (19).

Deep learning based models such as VGG19, convolutional neural networks (CNNs) and ResNet are highly applied because they automate the feature extraction from images that make them fast and dependable in identifying diseases. These models have been found to outperform other models in terms of their effectiveness in classifying potato leaf illnesses using conventional methods (20). Newer developments have brought with them hybrid methods that take the use of CNNs in conjunction with other algorithms, to improve the performance of disease classification. These models have shown significant performance on plant disease datasets such as potato leaf-oriented datasets with high performance evaluation measures (18). Moreover, such models must be incorporated into operational systems that can be accessed by smallholder farmers, necessitating easy to use interfaces that can operate in different field conditions. The application of ML and DL technologies in the diagnosis of potato leaf diseases could transform the agricultural practice to allow faster and more precise diagnoses, potentially allowing farmers to avoid crop losses and enhance productivity (19). However, existing studies show limitations including overfitting, poor generalisation and performances. Therefore, the present study proposes a hybrid ResNet50-BiLSTM model to improve performance of potato leaf disease classification. Detection and classification of potato leaf disease have turned out to be one of the research domains of serious concern because early and correct detection would help a great deal in reducing crop losses and enhancing yield. Through these innovative methods using hybrid models these approaches are not only improving the accuracy of the disease classification but are also laying the foundation to more sustainable agricultural practices.

Key contribution

The main objective of this study is to develop a hybrid DL model for accurate classification of potato leaf diseases using advanced preprocessing and deep feature extraction techniques. Our study also aims to demonstrate how the developed model can be used practically to support the timely prediction and classification of leaf illnesses in potato. The key contributions of the proposed work are summarised as:

1. Potato leaf images from the PlantVillage dataset were pre-processed by applying grayscale conversion, normalisation, resizing, denoising and bilateral filtering to improve disease related features.
2. A hybrid PhytoResLSTM model, integrating ResNet50-based feature extraction with bidirectional long short-term memory (BiLSTM) network, enabling effective learning of both spatial features and bidirectional dependencies to improve potato leaf disease classification.

The present study comprises the following sections. Section 2 involved a detailed analysis of various ML and DL techniques that are applied to predict and classify potato leaf diseases. Section 3 explains on about acquisition of leaf images from dataset, preprocessing of the images including extraction of features and classification of leaf diseases. Section 4 consists of approximately the results, discussion, graphic presentation of the findings and conclusions.

Present status of potato leaf disease classification

A comprehensive search was performed on multiple databases to understand the current status regarding the role of ML and DL methods to classify potato leaf diseases. The current studies give us a brief description of recent innovations in the field of plant leaf disease detection, exploring different methods, including advanced ML and DL techniques that were applied solely or in combined strategies to extract features and classify leaf diseases to enhance accuracy and efficiency. Researchers employed the VGG-19 pre-trained model for extraction of features and classification using artificial neural networks (ANN), SVM, KNN and LR. Logistic regression, a ML model obtained highest accuracy of 97.8 % on the testing set for leaf illness classification in potato (21). The leaf diseases were classified using 450 leaf images of potato from PlantVillage dataset (22). The researchers extracted the colour, texture and shape features using traditional methods including histogram, gray level co-occurrence matrix (GLCM) and Hu moments. Random forest (RF) model achieves an accuracy of 97.00 %, leading to automatic plant leaf disease detection. In another study, a total of 3000 leaf images were used for automatic classification of disease in potato (23). The colour, histogram and texture features were extracted and normalisation was applied using scaling techniques. The NaïveBayes (NB), KNN and SVM classifiers were utilised for leaf diseases classification. The findings revealed that SVM model showed the highest accuracy of 96.83 % in disease classification of potato.

Researchers also extracted the colour and texture features using traditional approaches including local binary pattern and histograms for the L^* , a^* and b^* components from the images. Among the different used methods, SVM model obtained the highest accuracy of 92.2 % for classification of leaf disease and 91.0 % for degree of infection classification (16). Previous researchers employed k-means clustering algorithm for image segmentation to separate the diseased and healthy leaves of potato (24). Feature were extracted using GLCM techniques and SVM model had attained test accuracy of 95.99 % for leaf disease classification in potato. The CNN and ResNet models evaluated for extraction of feature and classification of potato's leaf illness diagnosis using PlantVillage dataset (25). The researchers used 452 (30 %) images for testing and 1700 (70 %) images were used for model training. The modified model (ResNet) achieves accuracy of 99.62 % for

classification of diseases. In another study, researchers identified leaf diseases in potato and tomato using 6652 images. These pictures were split into three classes i.e. testing, validation and training in the portion of 20 : 20 : 60. The shape, edge, texture and colour features were extracted using operation of 2D convolution layer. The ResNet-9 outperformed the VGG-16 model with 99.25 % accuracy, 99.33 % recall, 99.67 % precision and 99.33 % F1-score (26). For feature extraction and detection of potato leaf diseases GoogLeNet and SAM-CNN (VGG19) were extensively utilised, which resulted into 97.68, 98.58, 98.42 and 98.21 % of precision, accuracy, recall and F1 score on diagnosis of the leaf disease in potato (27).

In another study, researchers utilised GLCM technique for extraction of features and SVM for potato's leaf illness classification. The suggested technique attained an accuracy of 87.17 % in leaf illness categorisation (28). A modified CNN with squeeze-excitation blocks and residual skip connections layer aimed to boost extraction of features using MobileNetV2 and classification using modified depth wise CNNs. This modified approach achieved 97.00, 98.00, 98.20 and 99.00 % of precision, accuracy, F1-score and recall, respectively (29). In other study, researchers extracted the shape, colour and texture features using conventional methods. The findings revealed that the RF model achieved an overall precision of 96.12 %, recall of 96.25 %, F1-score of 96.16 % and an accuracy of 95.99 % for potato leaf disease classification (30). The researchers also proposed a hybrid approach using VGG16 and XAI framework respectively, for the extraction of features and leaf disease categorisation in potato and obtained the overall 98.00 % accuracy (31). Overall, these studies highlight the evolving field of potato's leaf disease classification and highlights the architectural innovations and multimodal approaches using ML and DL techniques in improving the accuracy for predicting and classification of leaf disease in plants. However, many existing models rely solely on CNN architectures and do not explore hybrid approaches integrating deep feature extraction with sequential learning. Therefore, development of hybrid models combining CNN and BiLSTM may improve classification performance.

Proposed methodology

The methodology was proposed using PhytoResLSTM, a hybrid approach for classification of leaf disease in potato. The approach has been represented through a schematic flowchart that outlines the strategic process to classify the leaf images as healthy, late blight and early blight in potato (Fig. 1). The process begins with the acquisition of leaf images from the dataset, followed by several preprocessing steps including normalisation, denoising and filtering. The PhytoResLSTM is a hybrid approach that was developed by dense layers in which features were extracted by deep convolutional residual neural feature extraction unit (DCRNFE) technique using ResNet50. Bidirectional long short-term memory (BiLSTM) a DL model was extensively utilised to classify the leaf diseases of potato. BiLSTM model is a type of recurrent neural network (RNN) architecture, which works on data in both front and back directions. It works especially well with those tasks in which both previous and future inputs are relevant to the task. The model was trained using the Adam optimizer with a learning rate of 0.001, along with categorical cross-entropy as the loss function for multiclass classification. Further, standard mathematical equations were used to assess the leaf disease categorisation. Finally, the state-of-the-art comparison was carried out with the previous ML and DL models of feature extraction and categorisation of potato leaf diseases.

Acquisition of images from dataset

To achieve the research objectives, a standardised potato leaf disease dataset is sourced from the open-access PlantVillage dataset (32). This dataset is widely utilised in various artificial intelligence (AI) studies to develop automated plant leaf disease diagnosis models. The dataset contains a total of 2152 leaf images, including 1000, 1000 and 152 images of early blight, late blight and healthy leaves (0, 1 and 2 respectively). The leaf images of potato have been split into three subsets in the ratio of 70 : 20 : 10 to obtain the training, testing and validation sets. Table 1 shows the distribution of images in different classes along with training, validation and testing sets.

Table 1. Distribution of potato leaf images used for training, testing and validation

Class	Training (70 %)	Testing (20 %)	Validation (10 %)	Total images
Early blight	700	200	100	1000
Late blight	700	200	100	1000
Healthy	106	30	16	152
Total	1506	431	215	2152

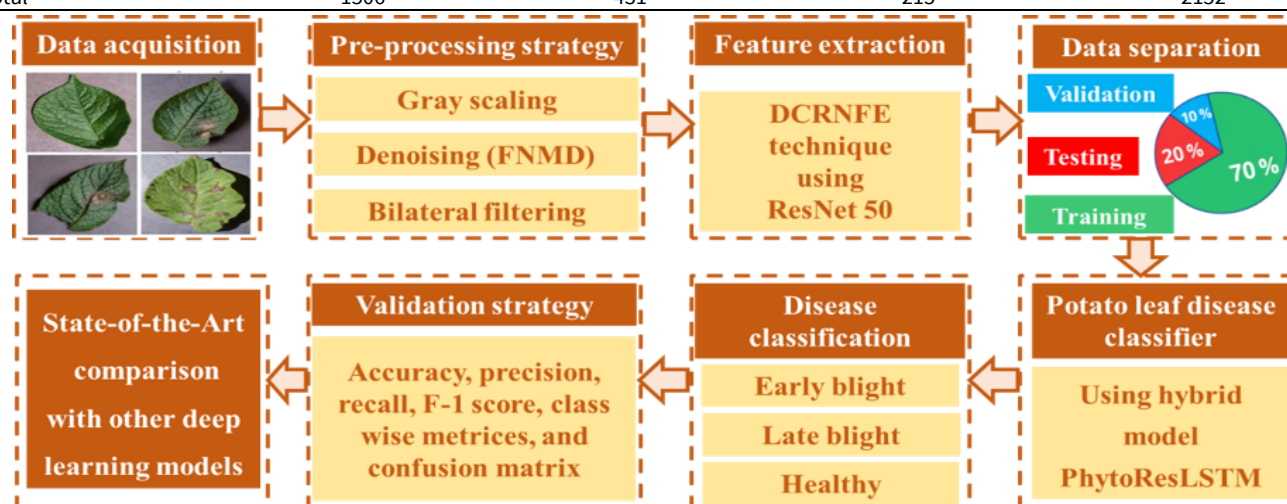


Fig. 1. Flow diagram of PhytoResLSTM model for potato leaf disease classification.

Image pre-processing

For accurate prediction and classification of leaf disease in potato, improving leaf image quality is an essential step and multiple techniques are available to standardise images before further steps to refine their details while maintaining essential disease-related characteristics. In pre-processing, as an initial step, all the images were converted to grayscale and normalised. It was utilised to convert the data into a standard normal distribution, assuring uniformity in the scale of all features. In this study, we converted the pixel intensities of images collected from the data source and normalised to obtain a standard deviation of 1 and a mean of 0 using equation (Eq. 1).

$$I_{\text{norm}}(x, y) = \frac{I(x, y) - \mu}{\sigma} \quad (\text{Eq.1})$$

In equation 1, $I_{\text{norm}}(x, y)$, $I(x, y)$, μ and σ presents the normalised intensity of pixels at coordinates (x, y) , original intensity of the pixel, mean of all pixel intensities and standard deviation of the pixel intensities of images. It reduces computational complexity, eliminates colour redundancy and enhances contrast between diseased and healthy leaf areas to make visible minor colour changes and spot patterns that represent a subtle disease related feature. After the normalisation, the images were resized to 64 x 64 pixels (width, height) using area interpolation to reduce the computations during classification.

In addition, the images were denoised to eliminate undesirable noise, distortions and irregularities caused by the external factors such as camera constraints, weather changes and varying lighting effects. Denoising was used to enhance the clarity of meaningful features of edges, patterns and disease symptoms, without any loss in significant structural and textual details of the images. In this work, the fast non-local means denoising (FNMD) method was employed to eliminate noise of normalised images (33). As presented in Fig. 2, the denoised images were further smoothed with bilateral filtering (34). Bilateral filtering is a weighted averaging method that is somewhat analogous to the Gaussian convolution, but with a crucial difference that enables it to maintain edges and smooth an image at the same time. Contrary to conventional smoothing filters that use the spatial proximity only, the bilateral filter considers the intensity difference between the neighbouring pixels. This means that a pixel influences another only if it is spatially close and has a similar intensity value. As a result, the bilateral filter effectively reduces noise in leaf images while maintaining lesion edges, vein structures and disease spots features that are crucial for accurate leaf disease analysis.

Feature extraction

The pre-processed pictures were then fed to the deep convolutional residual neural feature extraction (DCFRNE) unit for feature extraction using the ResNet50 architecture to obtain high-level representations from the 64 x 64 grayscale leaf images (Table 2). Since ResNet50 is inherently designed for three-channel RGB inputs, a preliminary 3 x 3 convolutional layers with three filters and ReLU activation was introduced to map the single-channel images to a 64 x 64 x 3 tensor without altering the spatial dimensions. This tensor was subsequently forwarded through the ResNet50 architecture initialised with pretrained ImageNet weights. Through fine-tuning, the network adapts to the attributes of potato leaf diseases while leveraging the rich representations acquired from extensive picture data. As the data progressed through the network, the feature maps were progressively down sampled via strided convolutional operations and pooling layers. Given the limited input resolution, the final convolutional layer of ResNet50 yielded a 2 x 2 x 2048 feature map. This output was then flattened into an 8192-dimensional vector, which served as a compact and discriminative representation of each image. The complete feature extraction model comprised approximately 23.6 million parameters (23.5 million trainable and 0.05 million non-trainable), the majority of which were contained within the ResNet50 backbone. For the dataset consisting of 2152 images, this procedure produced a feature matrix of size 2152 x 8192, which was subsequently used as input to the downstream classification module.

Leaf disease classification

In the present investigation, the leaf diseases of potato were classified using a RNN based DL model BiLSTM. This BiLSTM model consisted of two LSTM layers, the first layer processes the process sequence in the front direction and the following layer in the reverse direction. The outputs from both directions are typically concatenated or combined to form the final output, which is then passed through additional layers, such as dense or classification layers. Following feature extraction, the obtained class labels were reshaped into a column vector and encoded using one-hot encoding for the conversion of categorical class indices into a binary matrix representation. The encoding process produced a label matrix of shape (2152, 3), corresponding to three distinct classes of the image dataset. The image dataset was then divided into training, testing and validation subsets using a 70 : 20 : 10 split to ensure reproducibility.

The obtained feature vectors were fed to the DL model BiLSTM as shown in Fig. 3. Initially, the inputs were passed through three fully connected layers consisting of 2048, 1024 and 3200 units

Table 2. Feature extraction using ResNet50 from potato leaf images

Step No.	Layer (Type)	Description
1	Input initialisation	An input tensor with shape (H, W, 1) was defined to represent grayscale images. This prepared the data for the model.
2.	Channel expansion	Applied a 2D convolution with 3 filters of size 3 x 3, ReLU activation and 'same' padding to transform the single-channel grayscale images into 3-channel images in RGB format.
3.	Pretrained ResNet50 feature extraction	The 3-channel images were passed through the convolutional layers of ResNet50 pretrained on ImageNet. It excludes fully connected layers, producing hierarchical feature maps with spatial dimensions (2, 2, 2048) capturing high-level semantic features.
4.	Flatten feature maps	Converted multi-dimensional feature maps into 8192-dimensional 1D vectors using flattening to create a compact input image for further analysis.
5.	Model parameters	The approach employs 23,587,742 total parameters, including 23,534,622 trainable and 53,120 non-trainable parameters for feature extraction, which are subsequently input into BiLSTM, a DL model for leaf disease categorisation.

H: Image height; W: Image width; ReLU: Rectified Linear Unit activation function; RGB: Red-Green-Blue colour representation.

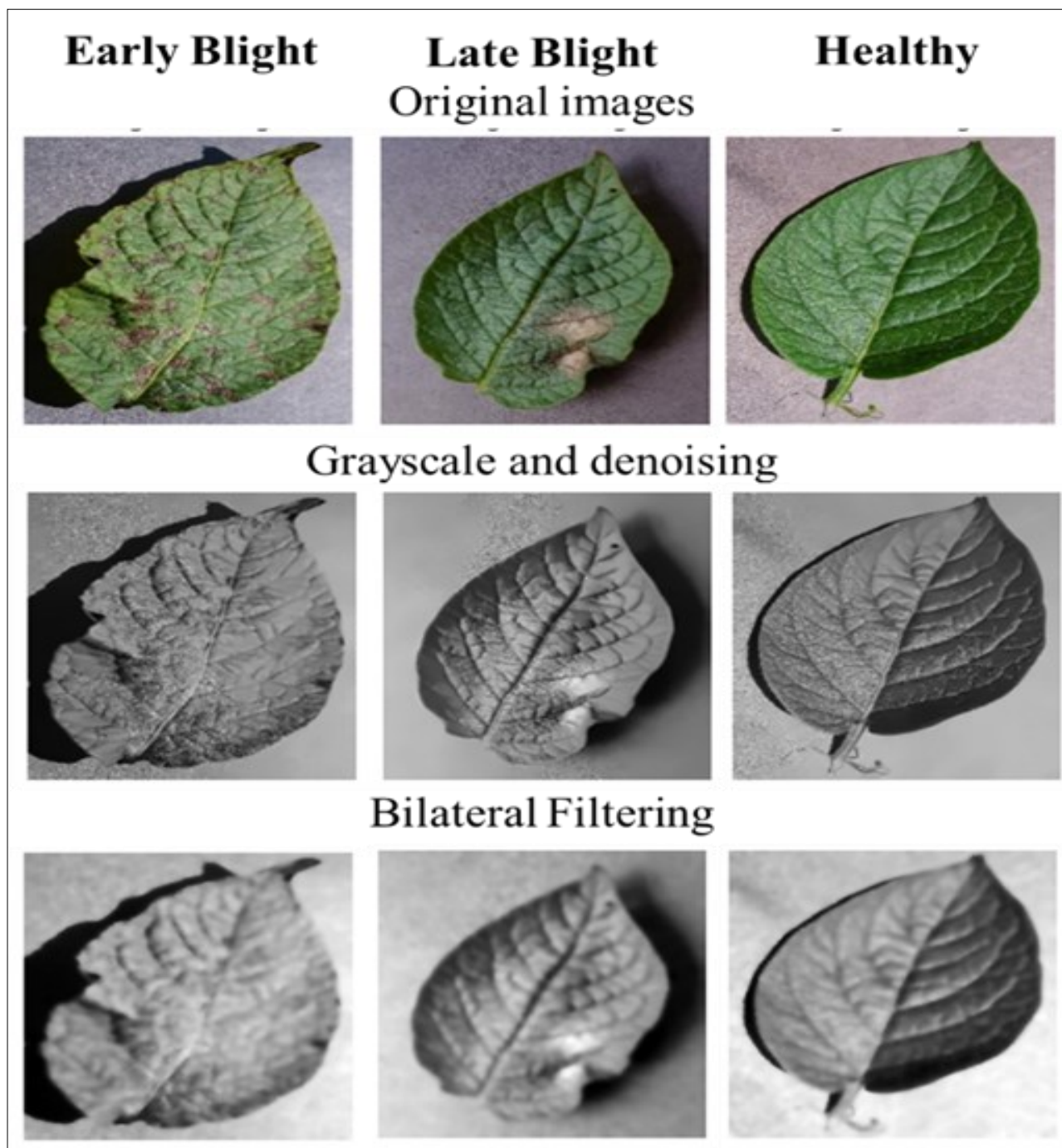


Fig 2. Pre-processing of the potato leaf images.

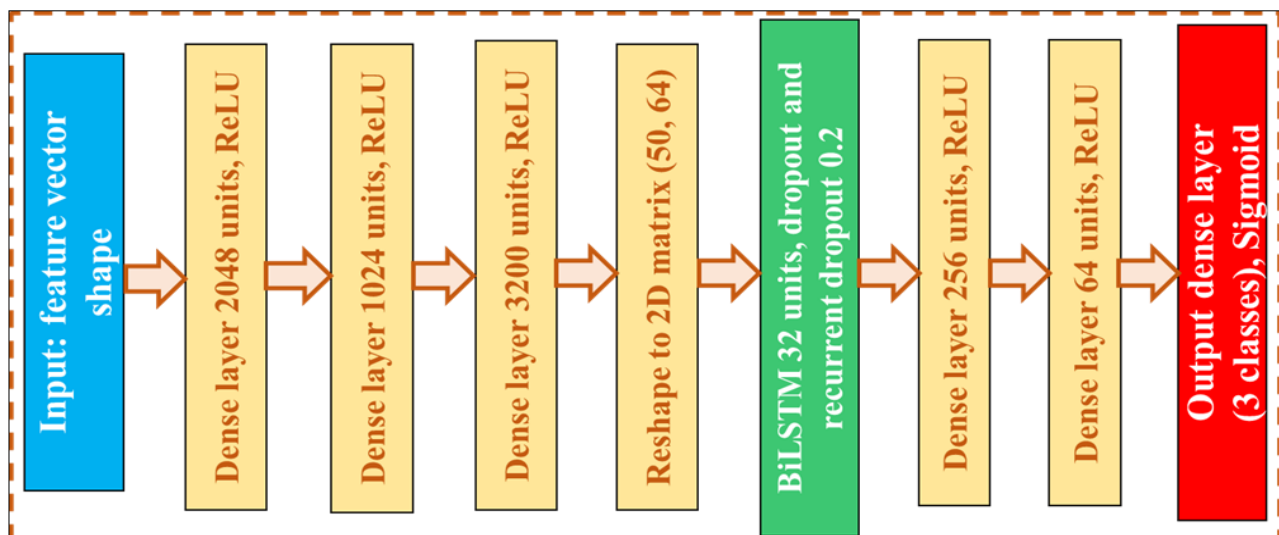


Fig. 3. The architecture of the BiLSTM model for potato leaf disease classification.

each adopting ReLU activation. The obtained units were then reshaped into a 2D sequence of length 50 with 64 features per time step and processed using BiLSTM layer with 32 units and dropout mechanisms. The multimodal feature of this LSTM layer enables the model to retain connections in both front and reverse routes, critical for detecting the sequential framework. The recurrent LSTM output was further passed through two additional dense layers with 256 and 64 units each adopting ReLU activation. Further filtration from dense output layer consisting of three neurons with sigmoid activation generated the multi-class leaf disease classification. The total number of trainable parameters in the model were 22, 215, 555.

Evaluation metrics

The proposed model PhytoResLSTM’s efficiency was evaluated using standard assessment metrics, including precision, accuracy, F1-score and recall, which are given in Table 3. The model’s performance was also evaluated using area under the curve (AUC) and receiver operating characteristic (ROC) curves, which illustrate the conflict between the false positive rate (FPR) and true positive rate (TPR) across varying decision thresholds. The confusion matrix shows the efficacy of the prediction model and the entries indicate how many predictions the model made in correctly or incorrectly recognised the classes.

Results and Discussion

Performance of PhytoResLSTM model

The performance evaluation of PhytoResLSTM includes quantitative class metrics, tracking training-validation loss and accuracy along with evaluation of confusion matrices and ROC curve interpretation. The performance of the proposed PhytoResLSTM model across various evaluation metrics including

accuracy, precision, recall and F1-score, according to class-wise setup is presented in Table 4. The precision for the early blight and late blight classes was nearly perfect at 1.00 and 0.9948 respectively, indicating that the approach is very efficient in predicting these illnesses. For healthy leaves, the precision (0.8438) was relatively lower, indicating that some healthy samples are incorrectly flagged as diseased (false positives), which may be influenced by the smaller sample size (28 instances). However, the high recall (0.9643) shows that the model rarely misses healthy leaves, successfully identifying most of them. Similarly, the very high recall for late blight (0.9794) and early blight (0.9952), showing that the model correctly identified most of the disease instances, with only a few misclassifications. The F1-score measures the harmonic mean of precision and recall, offers a balanced metric where false negatives and false positives are both significant. It gives a more comprehensive view of the model’s performance when there is a class imbalance, as is the case with healthy leaves, which has fewer instances in the dataset.

All classes obtained good F1-scores, particularly early blight and late blight with 0.98 and above, which represents balance between the precision and recall of these classes was near perfect. The F1-score of healthy leaves was comparatively less 0.90, which is still quite good taking into consideration the uneven distribution of classes. The less precision (0.8438) of this class also helps the F1-score to be lower since the predictions made by the model to the healthy leaves were not as accurate and more leaf misclassifications were made as diseased leaves in plants. Nevertheless, the recall of 0.9643 was high that means that most of the healthy cases were recognised correctly. The statistics of the macro average precision (0.9462), recall (0.9796) and F1-score (0.9615) were a bit less than that of the individual classes of early blight and late blight, which was the case of the imbalance in the dataset. The overall performance metrics in classifying the leaf

Table 3. Performance metrics

Metrics	Equation
Accuracy (A)	$A = \frac{TP^1 + TN^2}{TP + TN + FP^3 + FN^4}$
Precision (P)	$P = \frac{TP}{TP + FP}$
Recall (R)	$R = \frac{TP}{TP + FN}$
F-1 score (F)	$F = 2 \times \frac{(\text{Recall} \times \text{Precision})}{(\text{Recall} + \text{Precision})}$
TPR	$TPR = \frac{TP}{TP + FN}$
FPR	$FPR = \frac{FP}{FP + TN}$

¹TP: true positive; ²TN: true negative; ³FP: false positive; ⁴FN: false negative.

Table 4. Performance (%) for classifying leaf disease in potato

Classes	Precision	Recall	F1-score
Early blight	100.00	99.52	99.76
Late blight	99.48	97.94	98.70
Healthy	84.38	96.43	90.00
Accuracy			98.61
Macro average	94.62	97.96	96.15
Weighted average	98.75	98.61	98.65

diseases was decreased due to low performance metrics in healthy leaves. The weighted average precision (0.9875), recall (0.9861) and F1-score (0.9865) were however, more than the macro average values and showed better performance of the proposed model on the more common classes. The values imply that the larger classes were also able to significantly affect the predictions of the model, which is common with imbalanced datasets.

The multiclass ROC curve for the proposed approach trained to categorise various forms of potato leaf diseases are presented in Fig. 4. The ROC curve displays the TPR on the y-axis and the FPR on the x-axis. A higher TPR coupled with a lower FPR signifies superior model performance. The value of AUC varies from 0 to 1, with a value near to 1 signifying an enhanced performance of the model. The model attains an accurate classification for early blight, with an AUC value of 1.00, meaning it can perfectly distinguish early blight from the other classes. Like early blight, the model perfectly distinguishes late blight with an AUC value of 1.00. For healthy leaves, the model performs very well, with a high AUC value of 0.97, but slightly lower than the other classes. The diagonal line represents the performance of a random classifier, which classifies at random without learning from the data. At the final epoch (Epoch 8), the model attained an accuracy of 98.61 %, with a corresponding overall ROC-AUC of 0.9897, which represents that the proposed approach has excellent discriminatory power, correctly distinguishing between all three classes (early blight, late blight and healthy).

In the present study, the model was trained over 8 epochs and the performance metrics consistently improved throughout the training process. The training and validation loss and training and validation accuracy for the proposed approach is presented in Fig. 5. The loss was gradually declining as it was 0.6058 at the initial epoch and 0.1132 at the last epoch, indicating effective learning of the proposed model. At the same time, the categorical accuracy was increased with a high of 95.72 % on the 8th epoch, representing the model was learning to classify the potato disease categories as time progressed. Regarding validation performance, the validation loss was reduced to 0.0595 from 0.5211 and the validation accuracy improved from 81.90 % to 98.61 % in the first and last epoch, respectively. This signifies that the approach was quite predictive on previously viewed data and thus, there was less difference between the training and validation results and this is a good sign of the strength of the proposed model. As the categorical accuracy was high and the loss values were minimal at the final epoch, the model demonstrated excellent performance when classifying the potato leaf diseases, even when the model is tested with previously unknown data. The confusion matrix also supports the high level of performance of the model (Fig. 6.). The true positive values were 208, 190 and 27 cases of early blight, late blight and healthy leaf respectively, that were correctly identified. Misclassifications were minimal; for instance, only one early blight sample was misclassified as healthy, four late blight samples were misclassified as healthy and one healthy sample was misclassified as early blight. Such small errors indicate high reliability of the proposed approach in predicting and potato's leaf illnesses classification.

Table 5. State-of-art comparison of proposed model with recent leaf disease classification models

Reference	Crop	Dataset	Feature extraction	Classification	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
(35)	Potato	Publicly available	Not mentioned	VGG16	97.40	100.00	44.00	61.00
(36)	Potato	PlantVillage	DenseNet-201	DenseNet-201	97.20	98.09	99.28	98.67
(31)	Potato	PlantVillage	VGG16	XAI framework	98.00	98.00	98.00	98.00
(29)	Multiple crops including Potato	PlantVillage	MobileNetV2	DCNN with SE blocks and Residual Skip Connections	98.00	96.33	86.00	89.67
(37)	Three crops including Potato	Potato leaf disease dataset	Region adjacency graph (RAG)	GCN-GAT Hybrid architecture	98.11	98.11	98.11	98.11
(38)	Potato	PlantVillage	VGG19	Random forest	95.59	96.41	91.44	93.64
Proposed Model	Potato	PlantVillage	ResNet50	BiLSTM	98.61	94.62	97.96	96.15

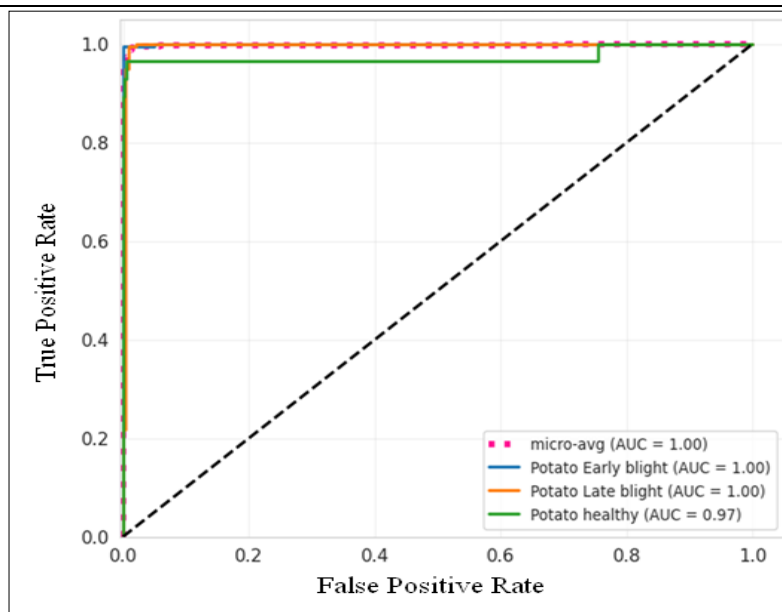


Fig. 4. Area under the receiver operating characteristic curve obtained from proposed model.

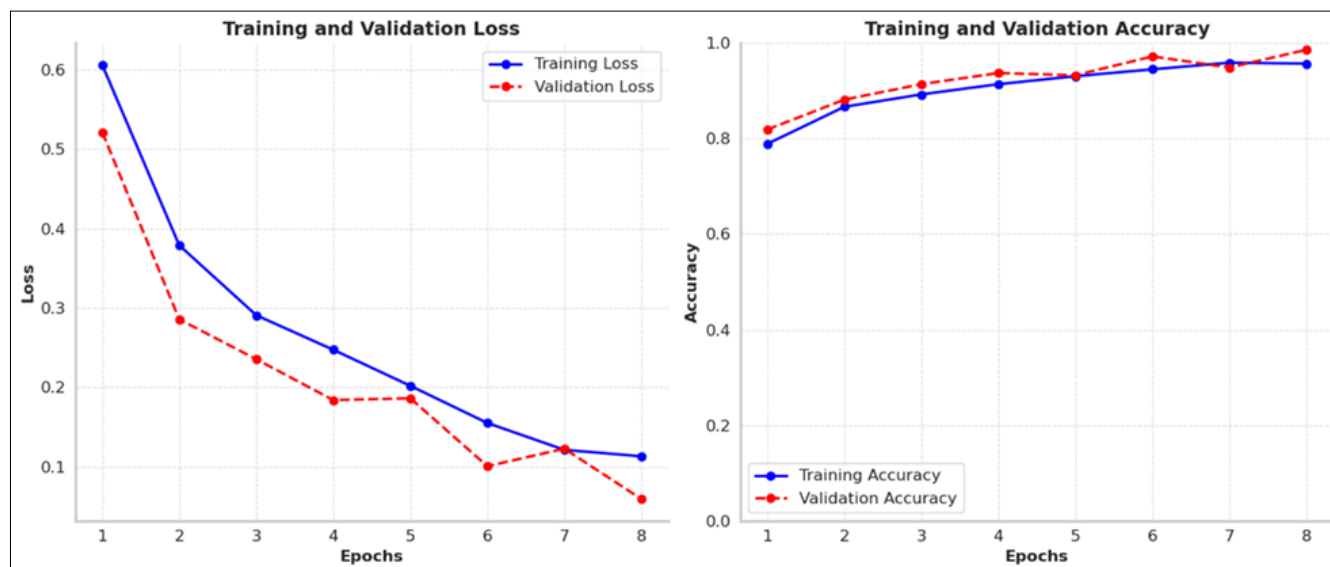


Fig. 5. Training loss and accuracy curves obtained from proposed model.

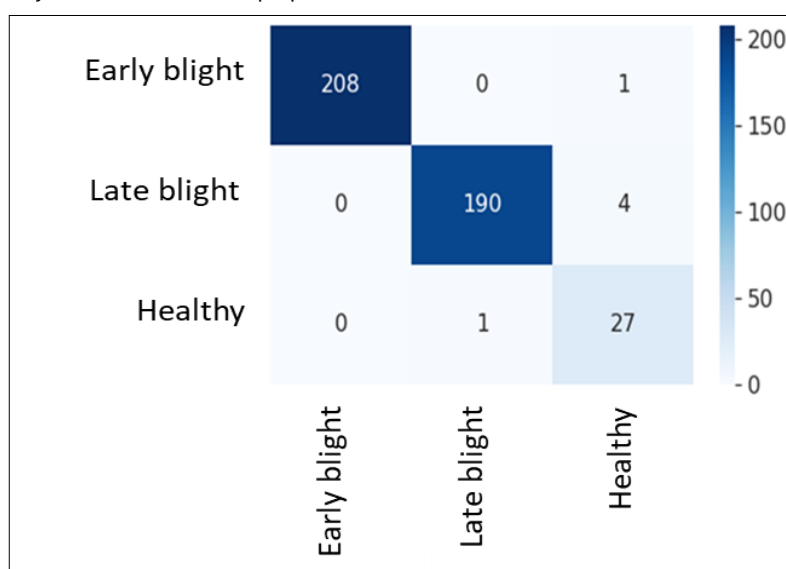


Fig. 6. Confusion matrix of leaf disease classification.

State of the art comparison

The performance of various models in potato leaf illnesses classification, including the proposed PhytoResLSTM model, is presented in the comparison table (Table 5). It is evident that current developments in machine learning (ML) and deep learning (DL) techniques have significantly improved the accuracy, precision, recall and F1-score of potato leaf disease detection models. In a study, VGG16 was employed for feature extraction and the accuracy of 97.40 % was achieved, with a precision (100 %) and a recall (44 %) demonstrating a disproportionate false positive and false negative ratio. This accuracy was ideal, but the model does not remember many true positive cases, hence cannot be utilised to a large extent in real life situations (35). Conversely, DenseNet-201 was used for feature extraction as well as potato leaf disease classification, achieved recall of 99.28 % and F1-score of 98.67 % and an accuracy of 97.20 % also demonstrating that the model was well-trained and the approach generalises well on the PlantVillage dataset (36). Subsequently, (31) applied VGG16 to extract features and an XAI framework and attained the accuracy of 98.00 % for leaf disease classification. Nevertheless, its relatively low precision of 98.00 % compared to the proposed model (98.61 %) and the absence of a more dynamic architecture can be a limitation to its capacity to adjust to more complex data sets or circumstances. In a more recent

study, used a GCN-GAT hybrid model with 98.11 % accuracy (37). These models are usually computationally expensive, which requires more resources and time to train and infer, making them less suitable for real-time and resource-constrained applications. For classification of potato leaf illness (38), employed a hybrid model PoLVR consisting of VGG-19 and RF for feature extraction and classification, in which 95.59 % accuracy was achieved.

The proposed hybrid model consisted of ResNet50 to extract features and BiLSTM to classify the potato leaf disease images, obtained incredible 98.61 % accuracy. The deep convolutional neural network ResNet50 extracted the low-level and high-level features from the image of potato leaves and contains complex patterns that show the presence of the disease. BiLSTM is a form of a recurrent neural network that enhances the models with the capability of processing and classifying a sequence of features and this makes the model especially useful in detecting subtle patterns of disease in various parts of the leaf. Training the model using the particular data will make sure that it adapts to the aspects of potato leaves thus optimising its performance. The balance between precision (94.62 %) and recall (97.96 %) illustrates its ability to decrease the number of false positives and false negatives and to identify diseased leaves at an early stage as well as to reduce

unnecessary treatments. The model with a good F1-score of 96.15 % has achieved a good trade-off between precision and recall. This demonstration highlights its practicality and usefulness, thus emerging as a potential answer in practical agricultural practice in which timely and accurate detection of disease is critical in ensuring the good health of crops and maximising their output.

Conclusion and future scope

In conclusion, the hybrid deep learning approach (PhytoResLSTM) shows its capability as a promising solution to the problem of potato leaf disease classification. The model is based on the high potential of the ResNet50 to extract features and the long-term dependency learning of the BiLSTM model for leaf illnesses classification. The model proves to be more efficient in classifying early blight, late blight and healthy potato leaves with the overall classification accuracy of 98.61 %. The evaluation metrics demonstrate that the model achieves strong and balanced performance across all classes. The close alignment between precision, recall and F1-score indicates effective learning with minimal bias toward any specific class. Furthermore, the high area under the curve (AUC) values reflect strong discriminative capability between classes, while the confusion matrix confirms reliable classification performance with limited misclassification, collectively supporting the robustness of the proposed approach. The comparison between the state-of-the-art reveals that the hybrid approach outperforms conventional approaches significantly, demonstrating the prospects that deep learning models have in the context of plant leaf disease detection. To further optimise the performance and applicability of the proposed hybrid approach, several research directions may be pursued in the future. The minor weakening in performance with the identification of healthy leaves indicates that the model can be further optimised to be more accurate in the process of identifying healthy and diseased classes. It will be possible to expand the dataset and incorporate a broader range of leaf diseases of plants and environmental setups to create a more generalised model. However, techniques like oversampling and class weighting were seen as useful future improvements to better handle class imbalance and enhance prediction accuracy. Moreover, the introduction of real-time data collection tools and integration of the model into mobile applications may enable farmers to promptly identify and regularly monitor the crops for the timely management of diseases, which may result in sustainable agricultural systems and food security.

Authors' contributions

ARM and PK contributed to conceptualisation and design of the study including writing the original draft of manuscript. ARM, PK and ST managed the literature searches. ARM and PK carried out the data processing and analysis, while SA reviewed and edited the paper. Each author participated in interpretation and revision of the manuscript. All authors have read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest: Authors do not have any conflict of interest to declare.

Ethical issues: None

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